Smrithi Lokesh Seramalanna

Email | Google Scholar | LinkedIn | Personal Website

Education

University of Pennsylvania, MSE Robotics **Expected: May 2027**

Birla Institute of Technology and Science, Pilani, B.E. (Hons.) Mechanical Engineering, Minor in Robotics and Automation – CGPA: 9.14/10.00 July 2025

Experience

Research Assistant, **Janus Intelligent Robot Lab, GRASP, University of Pennsylvania** – Philadelphia, Pennsylvania May 2026 – Present

- Built and maintain data-collection and data-handling pipelines for bimanual UR5 manipulation using GELLO teleoperation and wrist-mounted cameras; improved teleoperation reliability and standardized scene setup, demonstration collection, and post-processing.
- Train, fine-tune, and evaluate diffusion policies across approximately eight short-horizon and three long-horizon manipulation tasks, selecting checkpoints through repeated physical-robot deployments and failure analysis.
- Deploy Large Behavior Models trained from scratch and fine-tuned for long-horizon tasks lasting up to five minutes; debug failures across policy execution, UR controllers, and hardware integration.

Robotics Engineer, **Open Science Stack** – Bengaluru, India Aug 2025 – Dec 2025

- Developed and deployed a ROS 2 autonomy stack for xArm robots in an automated laboratory environment, integrating perception, motion planning, manipulation, and physical execution.
- Migrated the system from ROS and Gazebo to ROS 2 and NVIDIA Isaac Sim; integrated MoveIt2 with GPU-accelerated cuMotion and debugged sim-to-real, controller, planning, and hardware failures.

Exchange Student, **Mälardalens University** – Västerås, Sweden Jan 2025 - Jun 2025

- Recipient of European Union's Erasmus+ ICM Scholarship enrolled in advanced robotics coursework specializing in planning, perception and autonomous systems

Research Intern, **Carnegie Mellon University** – Pittsburgh, USA Jul 2024 - Jan 2025

- Developed a hierarchical planning framework for multi-robot collective construction, sequencing manipulation actions while verifying reachable robot access poses in dynamically changing environments.
- Extended Conflict-Based Search (CBS) for multi-robot navigation with carried-object, orientation, and collision constraints; validated the planner on 200 randomly generated structures.

Research Intern, **Indian Institute of Science x Cornell University** – Bangalore, India May 2024 - Jul 2024

- Researched soft robot stiffness in the Ruina Lab (Prof. Andy Ruina), designing and fabricating fiber-reinforced, pneumatically actuated soft bodies.
- Engineered an experimental setup to analyze how transmitted forces vary under controlled pressure, volume, and control-cable tensioning.

Research Intern, **Defence Research and Development Organisation** – Hyderabad, India May 2023 - Jul 2023

- Research in Advanced Systems Laboratory under Dr Lokesh Srivastava, Scientist 'F' on experimental characterization of carbon-epoxy composites for rocket motor casings and reliability prediction of composite samples by linear curve-fitting and Weibull analysis.

Publications

Hierarchical Planning for the Cuboid Multi-Agent Collective Construction (Cuboid-MACC) Problem Jun 2026

Junil Min, Shambhavi Singh, Geordan Gutow, ***Smrithi Lokesh Seramalanna***, Bhaskar Vundurthy, Jinwoo Choi, Howie Choset

Accepted to the 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Kinematic models for Cable-driven Continuum Robots with multiple segments and varying cable offsets Sept 2024

Ashish Bhalkikar, ***Smrithi Lokesh***, Ashwin K P

doi.org/10.1016/j.mechmachtheory.2024.105701

Projects

Autonomous VIO-Based Quadcopter Navigation – *MEAM 6200 Advanced Robotics* Spring 2026

- Integrated A* planning on a voxelized map, minimum-snap trajectory generation, SE(3) geometric control, and an Error-State Kalman Filter (ESKF) for IMU/stereo-visual fusion into a full closed-loop autonomy stack.
- Achieved safe navigation across six obstacle environments (maze, slalom, stairwell, switchback, window, over/under) using only on-board VIO state estimates; co-designed trajectory aggressiveness with estimator bandwidth to prevent visual feature dropout.

Two-View Stereo 3D Reconstruction – *CIS 5800 Machine Perception* Spring 2026

- Implemented a complete stereo pipeline: rectification, disparity map computation (SSD / SAD / ZNCC with L-R consistency), depth estimation, and back-projected point-cloud generation in world frame.
- Aggregated multi-pair reconstructions to produce dense 3D point clouds of real scenes.

Aircraft Design and Analysis September 2022 - April 2024

- Designed and built a micro-class fixed wing aircraft to demonstrate to demonstrate Urban Air Mobility (UAM) missions for AIAA's Design, Build and Fly 2024.

Software for Machine Component Design January 2024 - May 2024

- Developed a repository aimed at complementing the design process detailed in Shigley's Mechanical Engineering Design in collaboration with the Teaching Learning Centre at BITS Pilani. The repository served as a resource hub for mechanical engineering students and professionals.

Technical Skills

Robotics & Autonomy: Python, C++, ROS 2, MoveIt2, cuMotion, A*, CBS, ESKF/VIO, motion planning, state estimation, NVIDIA Isaac Sim, Gazebo, RViz

Robot Learning & Perception: PyTorch, behavioral cloning, diffusion policies, DAgger, GELLO teleoperation, OpenCV, stereo vision, multimodal robot data

Robots & Hardware: UR3e, UR5, xArm, Franka Emika Panda, Robotiq Grippers, wrist-mounted cameras, Arduino, Raspberry Pi

Design & Fabrication: SolidWorks, Fusion 360, 3D printing, laser cutting, CNC

Teaching and Volunteering

Graduate Teaching Assistant, **University of Pennsylvania** Aug 2026 – Present

- EAS 5160: Mathematical Foundations for Machine Learning I
- EAS 5170: Mathematical Foundations for Machine Learning II

Lab Staff, **Additive Manufacturing Lab (AddLab), University of Pennsylvania** May 2026 – Present

- Support research labs and student engineering projects through additive manufacturing, rapid prototyping, and fabrication.
- Operate and maintain industrial and desktop 3D printing systems, including FDM, SLA, and resin-based technologies.

Teaching Assistant – Artificial Intelligence, **Engineering Summer Academy, University of Pennsylvania** July 2026

- Working with Prof. Robert Ghrist, supporting high-school students in a summer program on foundational artificial intelligence and algorithmic thinking.

Scratch Instructor, **Fife-Penn STEM and CS Academy** – St. Francis Xavier School January 2026 – Present

- Teaching foundational programming concepts to K–8 students to prepare students for higher-level STEM education.

Undergraduate Teaching Assistant – Mechanics of Solids, **BITS Pilani** Aug 2023 – Dec 2023

- TA under Prof. Kiran Dinkar Mali; addressed student queries, evaluated quizzes, and managed academic records for a class of 180 sophomore students.

Awards and Achievements

Erasmus+ ICM Scholarship: European Union

Overall Excellence Award: International Rover Challenge 2023

26/107 International Rank: AIAA Design, Build and Fly 2024

Partial Travel Grant: International Programs and Collaborations Division, BITS Pilani